MaUWB ESP32S3 - Robot Board Schematic

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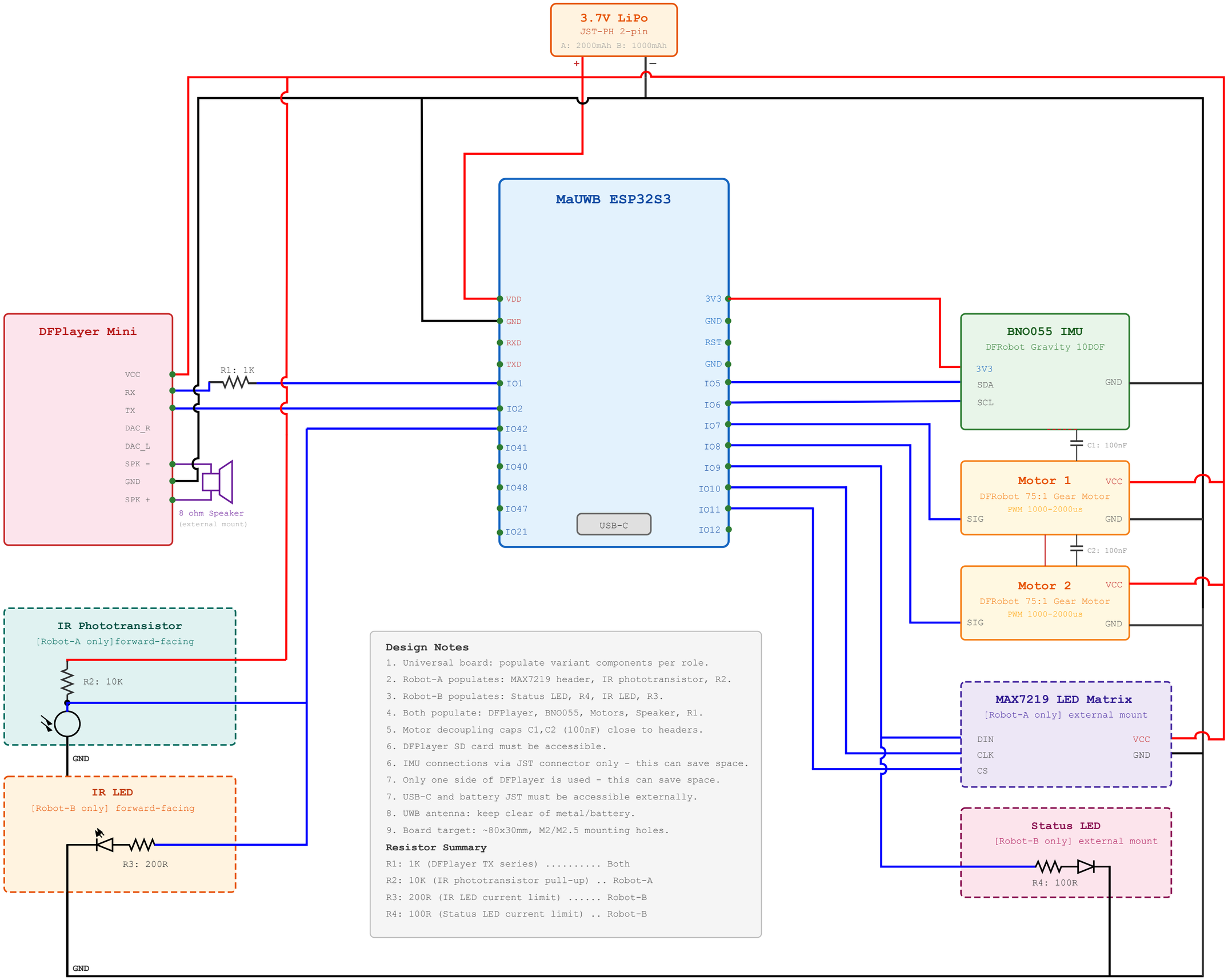

Design Notes

1. Universal board: populate variant components per role.
2. Robot-A populates: MAX7219 header, IR phototransistor, R2.
3. Robot-B populates: Status LED, R4, IR LED, R3.
4. Both populate: DFPlayer, BNO055, Motors, Speaker, R1.
5. Motor decoupling caps C1,C2 (100nF) close to headers.
6. DFPlayer SD card must be accessible.
6. IMU connections via JST connector only - this can save space.
7. Only one side of DFPlayer is used - this can save space.
7. USB-C and battery JST must be accessible externally.
8. UWB antenna: keep clear of metal/battery.
9. Board target: ~80x30mm, M2/M2.5 mounting holes.

Resistor Summary

R1: 1K (DFPlayer TX series) ..... Both
R2: 10K (IR phototransistor pull-up) .. Robot-A
R3: 200R (IR LED current limit) ..... Robot-B
R4: 100R (Status LED current limit) .. Robot-B